# Tutorial - 3

A retail Company wants to predict monthly sales revenue based on marketing & operational factors

Input features

→ Advertising Spend (₹ in thousands)
→ Number of Sales persons
→ store football (per month)
→ discount Rate (%)

| Adspend | Salesperson | football | discount | sales revenue |
|---|---|---|---|---|
| 50 | 5 | 1200 | 10 | 8.5 |
| 70 | 6 | 1500 | 12 | 10.2 |
| 40 | 4 | 1000 | 8 | 6.8 |
| 90 | 8 | 1900 | 15 | 14.0 |
| 60 | 6 | 1400 | 10 | 9.5 |
| 80 | 7 | 1700 | 14 | 12.5 |
| 55 | 5 | 1300 | 9 | 8.9 |
| 100 | 9 | 2100 | 18 | 16.0 |
| 65 | 6 | 1450 | 11 | 9.8 |
| 45 | 4 | 1100 | 8 | 7.2 |

1) define the dependent variables

dependent variable is monthly sales revenue because MSR depends on other factors like advertising spend number of salespersons, store football & discount free

2) Write the multiple linear regression model

multiple linear regression model

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

where $Y$ = Sales Revenue

$x_1$ = Advertising spend

$x_2$ = number of sales person

$x_3$ = store football

$x_4$ = discount rate

$\beta_0$ = intercept

$\beta_1, \beta_2, \beta_3, \beta_4$ = coefficients

3) Final model :-

Sales = $\beta_0 + \beta_1$ (Ad spend) + $\beta_2$ (sales persons) + $\beta_3$ (foot ball) + $\beta_4$ (discount)

Substitute Each row.

Row 1:- $8.5 = \beta_0 + 50\beta_1 + 15\beta_2 + 12000\beta_3 + 10\beta_4$

Row 2:- $10.2 = \beta_0 + 70\beta_1 + 4\beta_2 + 1500\beta_3 + 12\beta_4$

Row 3:- $6.8 = \beta_0 + 40\beta_1 + 4\beta_2 + 1000\beta_3 + 8\beta_4$

Row 4:- $14 = \beta_0 + 90\beta_1 + 8\beta_2 + 1900\beta_3 + 15\beta_4$

Row 5:- $9.5 = \beta_0 + 60\beta_1 + 6\beta_2 + 1400\beta_3 + 10\beta_4$

Row 6:- $12.5 = \beta_0 + 80\beta_1 + 7\beta_2 + 1700\beta_3 + 24\beta_4$

Row 7:- $8.9 = \beta_0 + 55\beta_1 + 5\beta_2 + 1300\beta_3 + 9\beta_4$

Row 8:- $16.0 = \beta_0 + 100\beta_1 + 9\beta_2 + 2100\beta_3 + 18\beta_4$

Row 9:- $9.8 = \beta_0 + 65\beta_1 + 6\beta_2 + 1400\beta_3 + 11\beta_4$

Row 10:- $4.2 = \beta_0 + 45\beta_1 + 4\beta_2 + 1100\beta_3 + 8\beta_4$

Dicuss the effect of discount rate on sales growth revenue

discount = 8, sales = 6.8, discount = 18,

Sales = 16, since $18 > 8 \Rightarrow 16 > 6.8$

4) Find MSE for 5 iterations:-

$$\boxed{MSE = \frac{1}{M}\sum_{i=1}^{N}(y - \hat{y}_i)^2}$$

$\hat{y}_i = [8, 9, 7, 13, 10, 11, 9, 15, 10, 8]$

Square Errors:-

$(0.5)^2 + (1.2)^2 + (-0.2)^2 + 1^2 + (-0.5)^2 + (-0.1)^2$

$+ 1^2 + (-0.2)^2 + (-0.8)^2$

$= 6.92$

$\Rightarrow MSE = \frac{6.92}{10} = 0.692$

Iteration - 2 :-

$\hat{y}_2 = [8.3, 9.8, 6.9, 13.6, 9.7, 12.8, 6, 15.5, 9.9, 7.5]$

Sum of Square Error = 2.01

$MSE_2 = \frac{2.01}{10} = 0.201$

Iteration - 3 :-

$\hat{y}_3 = [8.4, 10, 6.85, 13.8, 9.6, 12.3, 8.85, 15.3, 9.85, 7.35]$

$SSE = 0.62$

$MSE_3 = \frac{0.62}{10} = 0.062$

Iteration 4 :-

$\hat{y}_4 = [8.45, 10.1, 6.82, 13.9, 9.55, 12.4, 8.88, 15.9, 9.82, 7.28]$

$SSE = 0.18$

$MSE_4 = \frac{0.18}{10} = 0.018$

Iteration 5 :-

$\hat{y}_5 = [8.5, 10.21, 6.8, 14, 9.5, 12.5, 8.9, 16, 9.8, 7.2]$

$SSE = 0 \quad MSE = \frac{0}{10} = 0$